

Rosen — §4.4: Solving Congruences

Selected Exercises

Alston

Semester 2, 2026

Exercise 7

Show that if a and m are relatively prime positive integers, then the inverse of a modulo m is unique modulo m . [*Hint*: Assume that there are two solutions b and c of the congruence $ax \equiv 1 \pmod{m}$. Use Theorem 7 of Section 4.3 to show that $b \equiv c \pmod{m}$.]

Exercise 15

Show that if m is an integer greater than 1 and $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m/\gcd(c, m)}$.

Exercise 17

Show that if p is prime, the only solutions of $x^2 \equiv 1 \pmod{p}$ are integers x such that $x \equiv 1 \pmod{p}$ or $x \equiv -1 \pmod{p}$.

Exercise 18

- (a) Show that if p is a prime, the positive integers less than p , except 1 and $p-1$, can be split into $(p-3)/2$ pairs of integers such that each pair consists of integers that are inverses of each other. [*Hint*: Use the result of Exercise 17.]
- (b) From part (a) conclude that $(p-1)! \equiv -1 \pmod{p}$ whenever p is prime. This result is known as *Wilson's theorem*.
- (c) What can we conclude if n is a positive integer such that $(n-1)! \not\equiv -1 \pmod{n}$?

Exercise 19

This exercise outlines a proof of Fermat's little theorem.

- (a) Suppose that a is not divisible by the prime p . Show that no two of the integers $1 \cdot a, 2 \cdot a, \dots, (p-1)a$ are congruent modulo p .
- (b) Conclude from part (a) that the product of $1, 2, \dots, p-1$ is congruent modulo p to the product of $a, 2a, \dots, (p-1)a$. Use this to show that $(p-1)! \equiv a^{p-1}(p-1)! \pmod{p}$.
- (c) Use Theorem 7 of Section 4.3 to show from part (b) that $a^{p-1} \equiv 1 \pmod{p}$ if $p \nmid a$. [*Hint*: Use Lemma 3 of Section 4.3 to show that p does not divide $(p-1)!$ and then use Theorem 7 of Section 4.3. Alternatively, use Wilson's theorem from Exercise 18(b).]
- (d) Use part (c) to show that $a^p \equiv a \pmod{p}$ for all integers a .

Exercise 22

Solve the system of congruences $x \equiv 3 \pmod{6}$ and $x \equiv 4 \pmod{7}$ using the method of back substitution.

Exercise 26

Find all solutions, if any, to the system of congruences $x \equiv 5 \pmod{6}$, $x \equiv 3 \pmod{10}$, and $x \equiv 8 \pmod{15}$.

Exercise 29

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime integers greater than or equal to 2. Show that if $a \equiv b \pmod{m_i}$ for $i = 1, 2, \dots, n$, then $a \equiv b \pmod{m}$, where $m = m_1 m_2 \cdots m_n$. (This result will be used in Exercise 30 to prove the Chinese remainder theorem. Consequently, do not use the Chinese remainder theorem to prove it.)

Exercise 30

Complete the proof of the Chinese remainder theorem by showing that the simultaneous solution of a system of linear congruences modulo pairwise relatively prime moduli is unique modulo the product of these moduli. [Hint: Assume that x and y are two simultaneous solutions. Show that $m_i \mid x - y$ for all i . Using Exercise 29, conclude that $m = m_1 m_2 \cdots m_n \mid x - y$.]

Exercise 35

Use Fermat's little theorem to show that if p is prime and $p \nmid a$, then a^{p-2} is an inverse of a modulo p .

Exercise 40

Show with the help of Fermat's little theorem that if n is a positive integer, then 42 divides $n^7 - n$.

Exercise 41

Show that if p is an odd prime, then every divisor of the Mersenne number $2^p - 1$ is of the form $2kp + 1$, where k is a nonnegative integer. [Hint: Use Fermat's little theorem and Exercise 37 of Section 4.3.]

Exercise 44

Recall: let n be a positive integer and let $n - 1 = 2^s t$, where s is a nonnegative integer and t is an odd positive integer. We say that n passes Miller's test for the base b if either $b^t \equiv 1 \pmod{n}$ or $b^{2^j t} \equiv -1 \pmod{n}$ for some j with $0 \leq j \leq s - 1$.

Show that if n is prime and b is a positive integer with $n \nmid b$, then n passes Miller's test to the base b .

Exercise 48

Recall: a composite integer m is a Carmichael number if it is a pseudoprime to every base a relatively prime to it, that is, $a^{m-1} \equiv 1 \pmod{m}$ for every a with $\gcd(a, m) = 1$.

Show that if $n = p_1 p_2 \cdots p_k$, where $p_1, p_2, \dots, p_k$ are distinct primes that satisfy $p_j - 1 \mid n - 1$ for $j = 1, 2, \dots, k$, then n is a Carmichael number.

Exercise 54

Recall: a primitive root of a prime p is an integer r such that every integer not divisible by p is congruent to a power of r modulo p .

Show that 2 is a primitive root of 19.

Exercise 56

Let p be an odd prime and r a primitive root of p . Show that if a and b are positive integers in $\mathbb{Z}_p$, then

$$\log_r(ab) \equiv \log_r a + \log_r b \pmod{p-1}.$$

Exercise 58

Recall: an integer a is a quadratic residue of m if $\gcd(a, m) = 1$ and the congruence $x^2 \equiv a \pmod{m}$ has a solution.

Which integers are quadratic residues of 11?

Exercise 59

Show that if p is an odd prime and a is an integer not divisible by p , then the congruence $x^2 \equiv a \pmod{p}$ has either no solutions or exactly two incongruent solutions modulo p .

Exercise 62

Recall: if p is an odd prime and a is an integer not divisible by p , the Legendre symbol $\left(\frac{a}{p}\right)$ is defined to be 1 if a is a quadratic residue of p and -1 otherwise.

Prove Euler's criterion, which states that if p is an odd prime and a is a positive integer not divisible by p , then

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

[*Hint: If a is a quadratic residue modulo p , apply Fermat's little theorem; otherwise, apply Wilson's theorem, given in Exercise 18(b).*]

Exercise 64

Show that if p is an odd prime, then -1 is a quadratic residue of p if $p \equiv 1 \pmod{4}$, and -1 is not a quadratic residue of p if $p \equiv 3 \pmod{4}$. [*Hint: Use Exercise 62.*]